

**Kevin Cimorelli**  
Site Vice President

**Susquehanna Nuclear, LLC**  
769 Salem Boulevard  
Berwick, PA 18603  
Tel. 570.542.3795 Fax 570.542.1504  
Kevin.Cimorelli@TalenEnergy.com



November 3, 2022

Attn: Document Control Desk  
U. S. Nuclear Regulatory Commission  
Washington, DC 20555-0001

10 CFR 50.73

**SUSQUEHANNA STEAM ELECTRIC STATION  
LICENSEE EVENT REPORT 50-388/2021-004-01  
UNIT 2 LICENSE NO. NPF-22  
PLA-8000**

---

**Docket No. 50-388**

Attached is Licensee Event Report (LER) Supplement 50-388/2021-004-01. The LER supplement reports an event in which a Unit 2 Main Turbine Pressure Regulator was inoperable for longer than allowed by Technical Specifications (TS). The condition is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) as condition prohibited by TS.

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new or revised regulatory commitments.

A handwritten signature in black ink, appearing to read "K. Cimorelli", written over a horizontal line.

K. Cimorelli

Attachment: LER 50-388/2021-004-01

Copy: NRC Region I  
Mr. C. Highley, NRC Senior Resident Inspector  
Ms. A. Klett, NRC Project Manager  
Mr. M. Shields, PA DEP/BRP

**LICENSEE EVENT REPORT (LER)**

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form <https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to [Infocollections.Resource@nrc.gov](mailto:Infocollections.Resource@nrc.gov), and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk a/c: [oir\\_submission@omb.eop.gov](mailto:oir_submission@omb.eop.gov). The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |                                                                              |                      |                                                       |         |                                               |           |                                          |                                                    |               |       |     |      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------|----------------------|-------------------------------------------------------|---------|-----------------------------------------------|-----------|------------------------------------------|----------------------------------------------------|---------------|-------|-----|------|
| <b>1. Facility Name</b><br>Susquehanna Steam Electric Station, Unit 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |                                                                              |                      |                                                       |         | <b>2. Docket Number</b><br>05000388           |           |                                          | <b>3. Page</b><br>1 of 3                           |               |       |     |      |
| <b>4. Title</b><br>Unit 2 Main Turbine Pressure Regulator Inoperable for Longer Than Allowed by Technical Specifications due to Inadequate Valve Modification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |                                                                              |                      |                                                       |         |                                               |           |                                          |                                                    |               |       |     |      |
| <b>5. Event Date</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                              | <b>6. LER Number</b> |                                                       |         | <b>7. Report Date</b>                         |           |                                          | <b>8. Other Facilities Involved</b>                |               |       |     |      |
| Month                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Day    | Year                                                                         | Year                 | Sequential Number                                     | Rev No. | Month                                         | Day       | Year                                     | Facility Name                                      | Docket Number |       |     |      |
| 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 11     | 2021                                                                         | 2021                 | - 004 -                                               | 01      | 11                                            | 03        | 2022                                     | Facility Name                                      | Docket Number |       |     |      |
| 9. Operating Mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |                                                                              |                      |                                                       |         |                                               |           |                                          |                                                    |               |       |     |      |
| 10. Power Level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |                                                                              |                      |                                                       |         |                                               |           |                                          |                                                    |               |       |     |      |
| 11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |                                                                              |                      |                                                       |         |                                               |           |                                          |                                                    |               |       |     |      |
| 10 CFR Part 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        | <input type="checkbox"/> 20.2203(a)(2)(vi)                                   |                      | <input type="checkbox"/> 50.36(c)(2)                  |         | <input type="checkbox"/> 50.73(a)(2)(iv)(A)   |           | <input type="checkbox"/> 50.73(a)(2)(x)  |                                                    |               |       |     |      |
| <input type="checkbox"/> 20.2201(b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        | <input type="checkbox"/> 20.2203(a)(3)(i)                                    |                      | <input type="checkbox"/> 50.46(a)(3)(ii)              |         | <input type="checkbox"/> 50.73(a)(2)(v)(A)    |           | 10 CFR Part 73                           |                                                    |               |       |     |      |
| <input type="checkbox"/> 20.2201(d)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        | <input type="checkbox"/> 20.2203(a)(3)(ii)                                   |                      | <input type="checkbox"/> 50.69(g)                     |         | <input type="checkbox"/> 50.73(a)(2)(v)(B)    |           | <input type="checkbox"/> 73.71(a)(4)     |                                                    |               |       |     |      |
| <input type="checkbox"/> 20.2203(a)(1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        | <input type="checkbox"/> 20.2203(a)(4)                                       |                      | <input type="checkbox"/> 50.73(a)(2)(i)(A)            |         | <input type="checkbox"/> 50.73(a)(2)(v)(C)    |           | <input type="checkbox"/> 73.71(a)(5)     |                                                    |               |       |     |      |
| <input type="checkbox"/> 20.2203(a)(2)(i)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        | 10 CFR Part 21                                                               |                      | <input checked="" type="checkbox"/> 50.73(a)(2)(i)(B) |         | <input type="checkbox"/> 50.73(a)(2)(v)(D)    |           | <input type="checkbox"/> 73.77(a)(1)(i)  |                                                    |               |       |     |      |
| <input type="checkbox"/> 20.2203(a)(2)(ii)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        | <input type="checkbox"/> 21.2(c)                                             |                      | <input type="checkbox"/> 50.73(a)(2)(i)(C)            |         | <input type="checkbox"/> 50.73(a)(2)(vii)     |           | <input type="checkbox"/> 73.77(a)(2)(i)  |                                                    |               |       |     |      |
| <input type="checkbox"/> 20.2203(a)(2)(iii)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        | 10 CFR Part 50                                                               |                      | <input type="checkbox"/> 50.73(a)(2)(ii)(A)           |         | <input type="checkbox"/> 50.73(a)(2)(viii)(A) |           | <input type="checkbox"/> 73.77(a)(2)(ii) |                                                    |               |       |     |      |
| <input type="checkbox"/> 20.2203(a)(2)(iv)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        | <input type="checkbox"/> 50.36(c)(1)(i)(A)                                   |                      | <input type="checkbox"/> 50.73(a)(2)(ii)(B)           |         | <input type="checkbox"/> 50.73(a)(2)(viii)(B) |           |                                          |                                                    |               |       |     |      |
| <input type="checkbox"/> 20.2203(a)(2)(v)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        | <input type="checkbox"/> 50.36(c)(1)(ii)(A)                                  |                      | <input type="checkbox"/> 50.73(a)(2)(iii)             |         | <input type="checkbox"/> 50.73(a)(2)(ix)(A)   |           |                                          |                                                    |               |       |     |      |
| <input type="checkbox"/> Other (Specify here, in Abstract, or in NRC 366A).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |                                                                              |                      |                                                       |         |                                               |           |                                          |                                                    |               |       |     |      |
| 12. Licensee Contact for this LER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |                                                                              |                      |                                                       |         |                                               |           |                                          |                                                    |               |       |     |      |
| Licensee Contact<br>Peggy Kramer, Regulatory Affairs Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |                                                                              |                      |                                                       |         |                                               |           |                                          | Phone Number (Include Area Code)<br>(570) 542-3131 |               |       |     |      |
| 13. Complete One Line for each Component Failure Described in this Report                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |                                                                              |                      |                                                       |         |                                               |           |                                          |                                                    |               |       |     |      |
| Cause                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | System | Component                                                                    | Manufacturer         | Reportable to IRIS                                    | Cause   | System                                        | Component | Manufacturer                             | Reportable to IRIS                                 |               |       |     |      |
| B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | SB     | RTV                                                                          | B350                 | Y                                                     |         |                                               |           |                                          |                                                    |               |       |     |      |
| 14. Supplemental Report Expected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |                                                                              |                      |                                                       |         |                                               |           |                                          | 15. Expected Submission Date                       |               | Month | Day | Year |
| <input checked="" type="checkbox"/> No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        | <input type="checkbox"/> Yes (If yes, complete 15. Expected Submission Date) |                      |                                                       |         |                                               |           |                                          |                                                    |               |       |     |      |
| <b>16. Abstract</b> (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)<br>At approximately 13:21 on October 11, 2021, the Susquehanna Steam Electric Station, Unit 2, experienced an automatic reactor scram due to a main turbine trip. All turbine bypass valves (TBVs) opened, as expected, to control reactor pressure but did not subsequently re-close. Troubleshooting identified Pressure Transmitter PT-20101B as the cause of the TBVs failure to re-close. To control reactor pressure, operators manually closed the four main steam isolation valves and utilized the main steam line drains and the High Pressure Coolant Injection System. Further analyses showed the Pressure Transmitter PT-20101B failure was caused by failure of the pressure transmitter root valve resulting from main steam line vibrations created by the scram. The root valve was previously modified; however, station personnel did not validate that the modification would perform properly in all plant operating conditions. To correct the condition, the failed valve was replaced and applicable procedures were updated.<br><br>Based on failure information, it is likely that Pressure Transmitter PT-20101B was inoperable for longer than allowed by Technical Specification (TS) 3.7.8. Therefore, this event is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by TS.<br><br>There were no actual safety consequences associated with the described condition. |        |                                                                              |                      |                                                       |         |                                               |           |                                          |                                                    |               |       |     |      |

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to [Infocollects.Resource@nrc.gov](mailto:Infocollects.Resource@nrc.gov), and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk ail: [aira\\_submission@omb.eop.gov](mailto:aira_submission@omb.eop.gov). The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

| 1. FACILITY NAME                           | 2. DOCKET NUMBER | 3. LER NUMBER |                   |         |
|--------------------------------------------|------------------|---------------|-------------------|---------|
|                                            |                  | YEAR          | SEQUENTIAL NUMBER | REV NO. |
| Susquehanna Steam Electric Station, Unit 2 | 05000-388        | 2021          | - 004 -           | 01      |

**NARRATIVE****CONDITIONS PRIOR TO EVENT**

Unit 1 – Mode 1, approximately 100 percent Rated Thermal Power (RTP)

Unit 2 – Mode 1, approximately 95 percent RTP

At approximately 13:21 on October 11, 2021, the Susquehanna Steam Electric Station (SSES), Unit 2, experienced an automatic reactor scram due to a main turbine trip. Subsequently, all turbine bypass valves (TBVs) [EIS System/Component Codes: JI/PCV] immediately opened as expected.

**EVENT DESCRIPTION**

On October 11, 2021, at approximately 13:21, the Susquehanna Steam Electric Station Unit 2 reactor automatically scrambled due to a main turbine [TA/TRB] trip. The scram event is reported under LER-2021-003. The Unit 2 Control Room received indication of a main turbine trip with both divisions of the Reactor Protection System (RPS) [EIS System Code: JC] actuated and all control rods inserted. The TBVs opened automatically, as expected, to control reactor pressure but subsequently remained open. Reactor pressure was initially stabilized through manual closure of the Main Steam Isolation Valves (MSIVs) [SB/ISV] and maintained using the main steam line drains [SB/DRN] and the High Pressure Coolant Injection (HPCI) [BJ] System.

Based on failure information, it is likely that Pressure Transmitter PT-20101B was inoperable for longer than allowed by Technical Specification (TS) 3.7.8, "Main Turbine Pressure Regulation System." Therefore, this event is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by TS.

**CAUSE OF EVENT**

Previous to the scram, the root valve (2RV-PT20101B) [RTV] for the 'B' Mains Steam Pressure Secondary Transmitter, PT-20101B, experienced a steam leak on the valve packing. A temporary repair was installed to seal the leak using 2X sealant. During the Unit 2 scram vibrations from the main steam lines caused the 2X sealant to crack and separate from the valve stem. The valve subsequently closed resulting in sustained pressure indication to the TBVs, which remained open longer than required. In preparation of the temporary repair, the station did not validate that 2X sealant would perform properly under all plant operating conditions.

**ANALYSIS/SAFETY SIGNIFICANCE**

While failure of the pressure transmitter is a potential transient initiator when operating at full power, it is not credited in the mitigation of any accident. The TBVs are credited in multiple Updated Final Safety Analysis Report Chapter 15 analyses, all of which assume the bypass valves will open to mitigate the pressure transient from the event. As noted in the event summary, the failure of the pressure transmitter did not prevent the bypass valves from opening nor was there any other impact to components credited with mitigation. The condition described herein did not result in a safety system functional failure. Accordingly, this event will not be counted as a safety system functional failure under the Reactor Oversight Process Performance Indicators. There were no actual consequences to the health and safety of the public as a result of this event.

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to [Infocollects.Resource@nrc.gov](mailto:Infocollects.Resource@nrc.gov), and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk ail: [aira\\_submission@omb.eop.gov](mailto:aira_submission@omb.eop.gov). The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

**1. FACILITY NAME**

Susquehanna Steam Electric Station,  
Unit 2

**2. DOCKET NUMBER**

05000-388

**3. LER NUMBER**

| YEAR | SEQUENTIAL<br>NUMBER | REV<br>NO. |
|------|----------------------|------------|
| 2021 | - 004 -              | 01         |

**NARRATIVE****CORRECTIVE ACTIONS**

Key Corrective Actions Include:

1. Replace 2RV-PT20101B.
2. Lessons Learned were shared with applicable departments.
3. Revise procedures to restrict use of 2X sealant as applicable.

**COMPONENT FAILURE INFORMATION**

Component – EHC System Main Steam Pressure Transmitter Root Valve (2RV-PT20101B)

System – Main Steam

Manufacturer – Borg Warner

Part Number – 74630

**PREVIOUS OCCURRENCES**

None.